

A study on Post Quantum- Recent Techniques, Challenges and Bibliometric analysis

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Abstract— The advent of quantum computers spells doom for classical encryption schemes. Asymmetric key cryptosystems are easily solved by quantum computers using Shor's algorithm and Grover's algorithm in polynomial time compared to exponential time for classical brute forcing techniques. Therefore, current encryption schemes are easily decrypted by quantum computing technologies. This indicates a paradigm shift in the field of cryptography and necessitates the development of quantum resistant algorithms. Therefore, the field of post-quantum cryptography is vital to protect sensitive data, which includes grave matters of national security. Thus, this work highlights the need of accelerating the impactful research on quantum resistant cryptosystems along with a bibliometric study showcasing present research output. Moreover, throws light on the state-of-the-art quantum resistant algorithms along with their methodologies and challenges (in particular, the NIST finalists) and suggests future work needed to safeguard matters of public safety and national security.

Keywords— *Post quantum cryptography, quantum resistant algorithms, quantum computers, cryptosystems, CRYSTALS-Kyber, data security*

I. INTRODUCTION

The modern world relies massively on the seamless connectivity of digital communication systems. The internet alone creates and transmits 2.5 quintillion bytes of data every day and this constitutes only publicly available data [1]. To protect this data in transmission there have been major advances in encryption protocols. The current state of the art, RSA-4096, is impenetrable using classical computing if key generation is strong [2]. It is imperative that sensitive, confidential data related to matters of national security be encrypted securely against any and all threats.

The danger posed to classical encryption algorithms based on prime factoring, discrete log, and PKC (public key cryptography) by quantum computing is high but there are no quantum computers capable of breaking even the weakest encryption protocols as of 2024 [3]. However, future challenges and dangers do exist, quantum computing is a fast-developing discipline, and breakthroughs are to be expected which will put all digital communications at risk [4].

Shor's algorithm is a groundbreaking quantum algorithm which utilizes the properties of quantum computing to speed up integer factorization exponentially [5]. Therefore, even RSA-4096 can be cracked in a matter of a few hours as opposed to brute forcing on a classical computer which will take trillions of

years. The problem, though, is that no such quantum computing power exists. The largest factored number using Shor's algorithm is 21 [3].

SNDL, or Store-Now-Decrypt-Later (or HNDL) is an attack vector that banks on the development of quantum decryption which can crack current encryption protocols easily. Adversaries can intercept and store confidential data on their systems, to be decrypted later by capable quantum computers [6]. To solve this challenge the field of post-quantum cryptography has been developed. The goal of post-quantum cryptography is to develop cryptosystems that are secure against both quantum and classical computers and can interoperate with existing communications protocols and networks [7].

This attack vector is highly effective and poses a significant danger for national security data that remains critical for the foreseeable future such as the Aadhar Card details of Indian citizens and key military positions and other sensitive military data. Thus, Post-Quantum Cryptography is crucial for safeguarding such sensitive data.

This work highlights the research done in the field with a stress on the underlying methodology and future challenges along with suggestions for future work. It also underlies the need for PQC research for national security which include defence, finance and satellite communications. Algorithms have been analysed with respect to their quantum resistivity and underlying protocols. The latest algorithm and its shortcomings have been explored.

II. CRYPTOGRAPHY

Cryptography is a technique of securing communication by converting plain text into ciphertext. It involves various algorithms and protocols to ensure data confidentiality, integrity, authentication, and non-repudiation.

A. Modern Cryptography

Modern cryptography is based on mathematical functions called trapdoor functions which are easy to solve one way but impossible in the other, which means that only the recipient (who holds the key) can solve the problem, hence securely encrypting the data. There are two main ways this is achieved in the world of cryptography: Symmetric key cryptography and asymmetric key cryptography.

Symmetric key cryptography uses the same key for encrypting as well as decrypting the data. The state-of-the-art

implementation of symmetric key cryptography is the AES-256 (Advanced encryption standard). The obvious problem is the sharing of this key over an insecure channel.

Asymmetric cryptography, also known as public key cryptography, overcomes the insecure sharing problem by using two keys: a public key accessible to anyone and a secure private key, which only the recipient has. No operation can be performed without both the keys which means that only those with the private key can encrypt/decrypt the data.

PKC utilizes the trapdoor functions and problems such as integer factorization, discrete logarithm, and elliptic curves. The problem arises when the difficulty in solving these functions in the other way (factorizing integers in the case of RSA) vanishes, leading to the encryption being rendered obsolete.

B. Post-Quantum cryptography

The field of post-quantum cryptography aims to create encryption methods that are secure against both classical and quantum computing threats. This involves the development of new algorithms based on mathematical problems that are believed to be hard for quantum computers to solve.

TABLE I. CRYPTOGRAPHIC ALGORITHMS

Algorithm/ Protocol	Protocol class	Quantum resistant
CRYSTALS-Kyber	Lattice based	Yes
FALCON	Lattice based	Yes
SPHINCS+	Hash based	Yes
RSA	Integer factorization	No
AES	SP Network	No
Diffie Hellman	Elliptic curves	No
SIKE	Isogeny	Yes

Examples of some cryptographic algorithms are presented in Table 1. Examples include lattice-based cryptography, code based cryptography, hash-based cryptography, and multivariate polynomial cryptography [8]. These algorithms are designed to work with existing communication protocols and networks, ensuring a smooth transition as the threat of quantum computing becomes more imminent [7].

There are four main methodologies for post-quantum cryptosystems:

1. Code based cryptography relies on the hardness of solving certain problems in coding theory like Learning parity with noise [9]. It also uses error-correcting codes to replicate the trapdoor functions used in cryptography (i.e. make a problem easy to solve one way but extremely difficult the other). Classic McEliece cryptosystem is an example of this method.
2. Hash based cryptography relies on the security of cryptographic hashing functions like SHA-3. SPHINCS+ is a NIST finalist in the quantum algorithms competition based on the hashing method.
3. Multivariate polynomial cryptography relies on multiple polynomial functions over a finite fields; which are known to be NP complete. Thus these are good contenders for post-quantum cryptosystems.

4. Lattice based cryptography relies on the mathematical field of lattice theory which contains problems like the shortest vector problem useful for cryptographic encryption techniques. CRYSTALS-Kyber and CRYSTALS-Dilithium are two examples of lattice based cryptosystems.

III. QUANTUM RESISTANT ALGORITHMS

As the development of quantum computers progresses, the need for post-quantum cryptographic solutions becomes more pressing. The National Institute of Standards and Technology (NIST) has been actively involved in the standardization of post-quantum cryptographic algorithms, with recent drafts specifying algorithms derived from CRYSTALS-Dilithium, CRYSTALS-KYBER, and SPHINCS+ [7]. Researchers and security firms worldwide are racing to develop and implement these new approaches to cryptography, aiming to protect data and communications against future quantum attacks.

We discuss the general purpose algorithm CRYSTALS-Kyber in this section. This algorithm is useful for data in transit which is important to protect since symmetric key encryption is hard.

A. CRYSTALS-KYBER

Bos et al. have introduced CRYSTALS-Kyber, a module-lattice-based Key Encapsulation Mechanism (KEM), as part of the broader CRYSTALS cryptographic suite [10]. The research focuses on constructing a KEM that is secure against quantum computer attacks, beginning with a CPA-secure public-key encryption scheme and then enhancing it with a variant of the Fujisaki-Okamoto transform to achieve CCA security. This security is robust against adaptive chosen-ciphertext attacks, which are among the most powerful types of attacks known [11].

CRYSTALS-Kyber is presented as an evolution of the NewHope KEM, offering several improvements. Notably, it features reduced key and ciphertext sizes—approximately half that of NewHope's. Moreover, Kyber provides CCA security, an upgrade from NewHope's passive security, and bases its security on a more general lattice problem. These enhancements do not compromise efficiency; Kyber's optimized implementation runs in a time frame comparable to NewHope.

The foundation of Kyber's security lies in the hardness of the Module-LWE problem, which is assumed to be difficult for both classical and quantum computers [12]. The authors set parameters aiming to provide a conservative security level of over 128 bits of post-quantum security, preparing for the advent of quantum computing that could potentially compromise current cryptographic systems.

In essence, Bos et al.'s work details the design and efficacy of CRYSTALS-Kyber within the CRYSTALS suite, emphasizing its potential as a quantum-resistant cryptographic solution [10].

This algorithm is regarded as one of the best post-quantum algorithms for general purpose encryption/decryption, but there are a few vulnerabilities. Theoretically, Elias et al. have shown instances where Ring-LWE based cryptography is weak [13]. Besides, this algorithm is also susceptible to side channel attacks; discussed in the next section.

B. Shortcomings

Lavich et al. conclude in their paper that the suggested key encapsulation system, CRYSTALS-Kyber, is confronting increasing difficulties due to advanced side-channel attacks. Current studies reveal weaknesses even in cases with strong security, challenging ongoing defensive advancements. Masking and shuffling are two strategies required for securing cryptosystems. The need to check algorithms for both mathematical strength and toughness to outside attacks increases as we approach the post-quantum era.

Side-channel attacks rely on unconventional ways to deduce the algorithm using data from methods like power usage analysis and time. This data is fed to an oracle which analyzes the data and gives the key to the attacker. However, as concluded by Lavich et al., this method relies on access to the hardware and there are ways to mitigate it.

One such way is known as masking. It entails dividing a hidden value into several parts randomly which are independent of each other and then operations are performed on them. This method is useful for mitigating electromagnetic and power analysis side channel attacks [14].

IV. DISCUSSION

Evaluation metrics for any new post-quantum encryption protocols should involve their strength against straightforward cryptographic attacks; quantum attacks; computational efficiency and ease of implementation along with robust defenses against futuristic side-channel attacks involving oracles supported with mathematical validation. The oracle threat is based on Artificial Intelligence. It basically entails a machine that monitors the data encryption process (this might mean gaining physical access to the machine doing the encryption and it might also be possible to do this remotely) and learns certain patterns that might reveal the encryption key and process.

A. Importance of Post-quantum cryptography

Mosca's inequality states that if the time until quantum computing reaches decrypting capabilities is greater than or equal to the amount of time our data needs to be secured for plus the time it takes to migrate to PQC encryption, then time has already run out [15].

$$D_T + A_T \geq Q_T \quad (1)$$

Here D_T is the amount of time our data needs to be kept secure, A_T is the amount of time needed for adoption of quantum resistant cryptosystems into existing networks and Q_T is the amount of time till quantum computing is at a level where it can crack current encryption protocols.

This inequality, if going by the progress of quantum computing, is close to being reached and the SNDL attack method also reduces the time limit to act. This justifies the need for rapid adoption of PQC protocols into existing networks, especially ones related to critically important matters of national security.

Due to this, NIST launched its quantum cryptography competition in 2017, with the round three finalists being based

on mathematical underpinnings of lattices, hashes and multivariate functions.

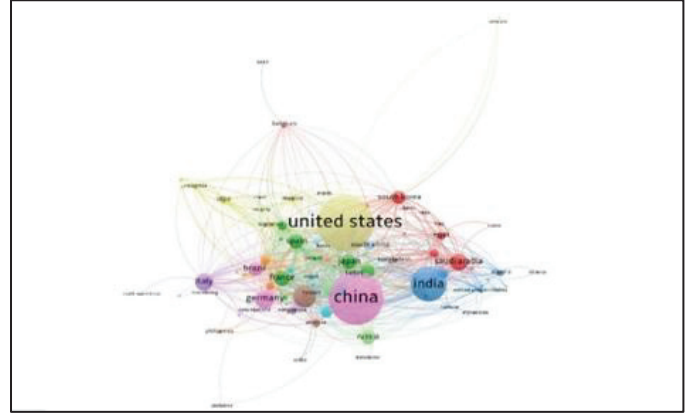


Fig. 1. Bibliometric country map on Quantum Resistant Algorithms

Figure 1. shows the research done on QRAs worldwide. Apart from USA, China and to some extent, India, there is a lack of research done on this problem worldwide. There is a need for a quantum leap of research output in the field.

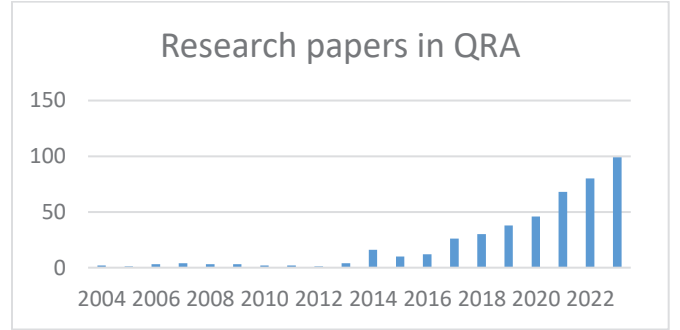


Fig. 2. Research papers published yearly in Quantum Resistant Algorithms

Fig 2. Shows the massive upward trend in research output in the field of quantum resistant algorithms over the last twenty years, signifying its importance as a crucial field.

Fig 1 and Fig 2 are obtained by carrying out bibliographic analysis by considering the research publication data on Quantum resistant algorithms from Scopus database.

B. Future work and suggestions

Future work in PQC must focus on adoption of the quantum resistant algorithms by the modern network infrastructure, i.e. how to migrate to quantum resistant algorithms. Challenges faced in adoption of quantum resistant algorithms are based on their novelty and the fact that adapting them to protect critical data right off the bat is unfeasible. Thus, we suggest that they be adopted *on top of* classical state-of-the-art encryption systems to mitigate the challenges brought about by the novelty of the algorithms.

We give a few suggestions on future direction of research:

- There is a need to focus on hybrid adoption i.e. using current protocols along with quantum proof encryption to protect data against SNDL attacks and to also get rid

of the drawbacks of early adoption like side channel attacks on the quantum resistant cryptosystems.

- Work needs to be done on mitigating side channel attacks by using techniques like masking to protect against all attack vectors.
- Collaboration between the research community and the open-source community can result in rapid prototyping of the existing PQC protocols and hence speed up the adoption of quantum resistant algorithms.

Roadmap for QRA Adoption:

1. Development of indigenous (in-house) quantum resistant algorithms
2. Testing these algorithms on non-critical datasets and against main and side channel attacks
3. Deployment based on established software development practices (i.e. rolling deployment, A/B testing, monitoring, rollback strategies and CI/CD)
4. Continued research on future proofing these algorithms

V. CONCLUSION

Quantum computing is the next revolution in the technological advancement of humanity. Due to the laws of physics, existing transistors are already hitting the minimum size limit. All of this points to quantum computing developing even faster than before. This, however, is not without downsides. There is a race brewing; between quantum computing aiming to break current encryption standards thus gaining access to all the data in the world, and post-quantum cryptography which aims to secure this data even before such quantum computers are realized.

What adversaries can do right now is harvest data to decrypt it later, while PQC can create quantum resistant algorithms and secure that data right now. Therefore, even if the speedups of quantum decryption are in the future, the race is in full swing right now.

This work has highlighted the importance of quantum cryptography for national security, and the need for more research into the topic with suggestions on where that research might benefit society the most.

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